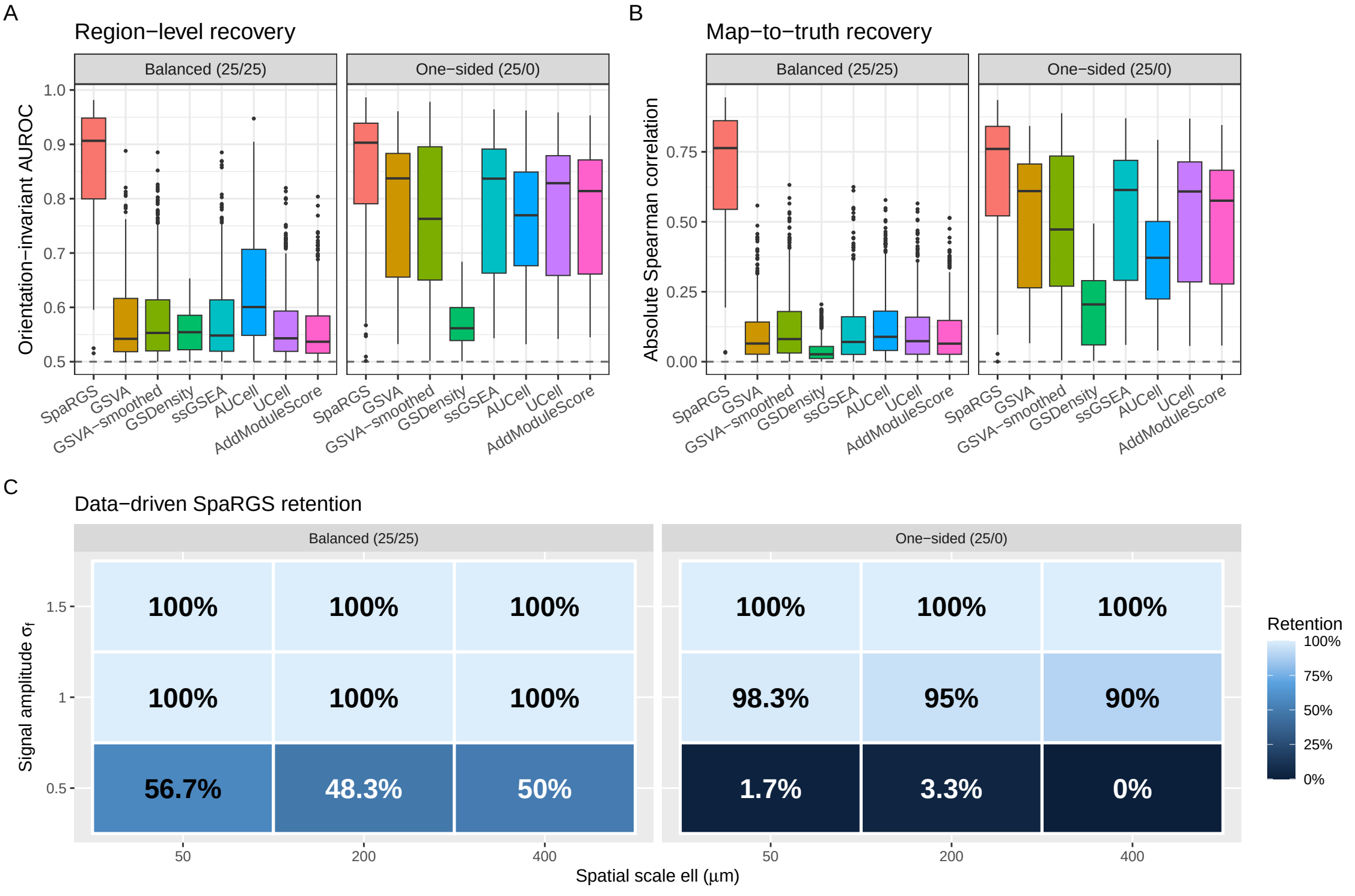
A
Region−level recovery
Balanced (25/25)
One−sided (25/0)
Orientation−invariant AUROC
1.0
0.9
0.8
0.7
0.6
0.5
SpaRGS
GSVA
GSVA−smoothed
GSDensity
ssGSEA
AUCell
UCell
AddModuleScore
B
Map−to−truth recovery
Balanced (25/25)
One−sided (25/0)
Absolute Spearman correlation
0.75
0.50
0.25
0.00
SpaRGS
GSVA
GSVA−smoothed
GSDensity
ssGSEA
AUCell
UCell
AddModuleScore
C
Data−driven SpaRGS retention
Balanced (25/25)
One−sided (25/0)
Signal amplitude σf
1.5
1
0.5
100%
100%
100%
100%
100%
100%
56.7%
48.3%
50%
100%
100%
100%
98.3%
95%
90%
1.7%
3.3%
0%
50
200
400
Spatial scale ell (μm)
Retention
100%
75%
50%
25%
0%